# Hritwik Biyani

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## EDUCATION

**Stony Brook University** — **Stony Brook, New York**
Master of Science in Computer Science (GPA: 4.0/4.0) — *August 2025 – August 2027*
*Relevant Coursework: CSE 535 (Distributed Systems), CSE 506 (Operating Systems)*

**MIT World Peace University** — **Pune, India**
Bachelor of Technology in Computer Science — *July 2017 – July 2021*

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, TypeScript, JavaScript, C, Bash, GraphQL, SQL, Distalgo
**Distributed Systems:** Paxos, PBFT, 2PC, Consensus Protocols, Sharding, Replication, Linearizability, Benchmarking
**Skills:** Data Structures, Microservices, CI/CD, REST APIs, Concurrency, Operating Systems, Networking
**Tools / Frameworks:** Git, React, AWS, Kafka, Spring Boot, Snowflake, Databricks, Node.js, Flask, Neo4j, Docker

## WORK EXPERIENCE

**Intuit** — **New York City**
*Software Engineer Intern, QuickBooks Online* — *June 2026 – August 2027*

- Developed a real-time tray and email-based notification service for Quickbooks task management system in Java with AWS DynamoDB, SQS, EventBridge and Numaflow - serving ~300K users in the pilot
- Implemented a major UI enhancement to simplify and speed up user experience for task updates - converted from drawer popup to inline editing using React and TypeScript, deployed to ~1.9M users
- Setup data pipelines and observability to aid in latency improvement of a quick action recommendation model (LightGBM) by four times - p95 latency went from ~640ms to ~140ms

**ZS Associates** — **Pune, India**
*Senior Data Engineer* — *July 2023 – July 2025*

- Engineered a fault-tolerant ingestion service integrating 40+ heterogeneous sources (REST APIs, FTP, SQL) with structured error logging and retry logic, reducing time-to-production for new data sources by 4–5 weeks.
- Led a team of 4 to design an event-driven data governance system using serverless functions and asynchronous triggers, with approval workflows and audit logging for data change requests automating away manual data entry
- Optimized query execution and schema design during a large-scale platform migration, restructuring data layouts to reduce compute overhead and improve backend latency by ~25%.

**ZS Associates** — **Pune, India**
*Data Engineer* — *November 2020 – June 2023*

- Built a Node.js backend service for real-time contracting scenario analysis, processing tens of thousands of rows via configurable algorithms to optimize contract selection and boost ROI by 30%+.
- Developed a lazy-loaded graph visualization tool using D3.js and Neo4j, writing optimized Cypher queries to efficiently render complex health system hierarchies with traversals across 10,000+ nodes.

## PROJECTS

**Constraint Solving in Python** | *Stony Brook University* — **February 2026 - *present***

- Wrote a degree planning constraint solver in Python using ORTools and docplex to compare with other approaches (prolog, clingo). Automated testing and benchmarking of different approaches in Python

**SBUnix Operating System Kernel** | *Stony Brook University* — **February 2026 - *present***

- Writing an operating system kernel from scratch in C and Assembly. Worked on the file systems development, terminal and memory allocators, also wrote a custom micropython port that works with our OS.

**Sharded Transactional Database (2PC + Paxos)** | *Stony Brook University* — **October 2025 - December 2025**

- Built a distributed, key-value store in Java with cross-shard ACID transactions with linearizable consistency.
- Implemented Multi-Paxos replication with cluster reconfiguration, and integrated 2-Phase Commit
- Achieved ~5,000 TPS under concurrent workloads using fine-grained row-level locking